

Naive Bayes Foundations: Mutually Exclusive Events

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Conditional probability ([Note MA-015](#)).
- **Leads to:** Naive Bayes ([Note ML-081](#)); Voting ensembles ([Note ML-096](#)).

1. Overview

Key point: Two events are mutually exclusive when they cannot happen together: $P(A \cap B) = 0$. Mutual exclusivity is a different idea from independence, and the two are easily confused.

The previous Note defined **independent** events (G-934): they can happen together, but one does not affect the other. **Mutually exclusive** events are different: they can never happen together. The two ideas are easy to mix up, so this Note puts them side by side.

2. The definition

Key point: Mutually exclusive: $P(A \cap B) = 0$, so $P(A | B) = 0$.

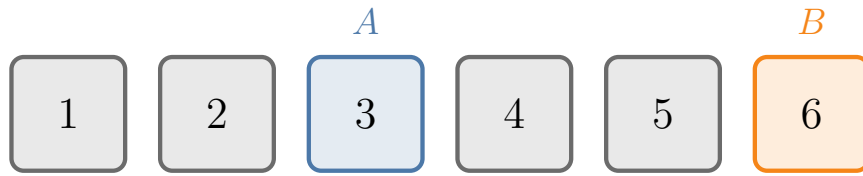
Some pairs of events can never happen together:

- A single coin toss cannot land both heads and tails.
- At a junction, a driver turns either left or right, not both.
- One die cannot show both 3 and 6 on the same roll.

Such events are **mutually exclusive** (G-1286): they have no outcome in common, so at most one of them can happen. As a formula, A and B are mutually exclusive if

$$P(A \cap B) = 0$$

one roll of one die



no face is in both: $P(A \cap B) = 0$

given B , only face 6 is left: $P(A \mid B) = 0$

A or B : 2 of 6 faces, $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

Figure 1: One roll of one die, with A = “shows 3” and B = “shows 6”. The two events share no face

Figure 1 shows the die example: A and B sit on different faces, so no outcome is in both. Once we know B happened, the roll is face 6, and face 3 is ruled out.

3. Conditional probability for mutually exclusive events

Key point: If B happened, A cannot have happened: $P(A \mid B) = 0$.

Putting $P(A \cap B) = 0$ into the conditional probability formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0}{P(B)} = 0$$

With numbers: die 1 has shown 3. The probability that the same die shows 6 on that roll is 0. A driver has turned left; the probability that they turned right is 0.

Figure 2 checks this on a million simulated rolls. Over all rolls, a 3 comes up 0.1667 of the time. Among the rolls that showed 6, not one showed 3: the share is exactly 0. The right panel previews the addition rule of Section 5.

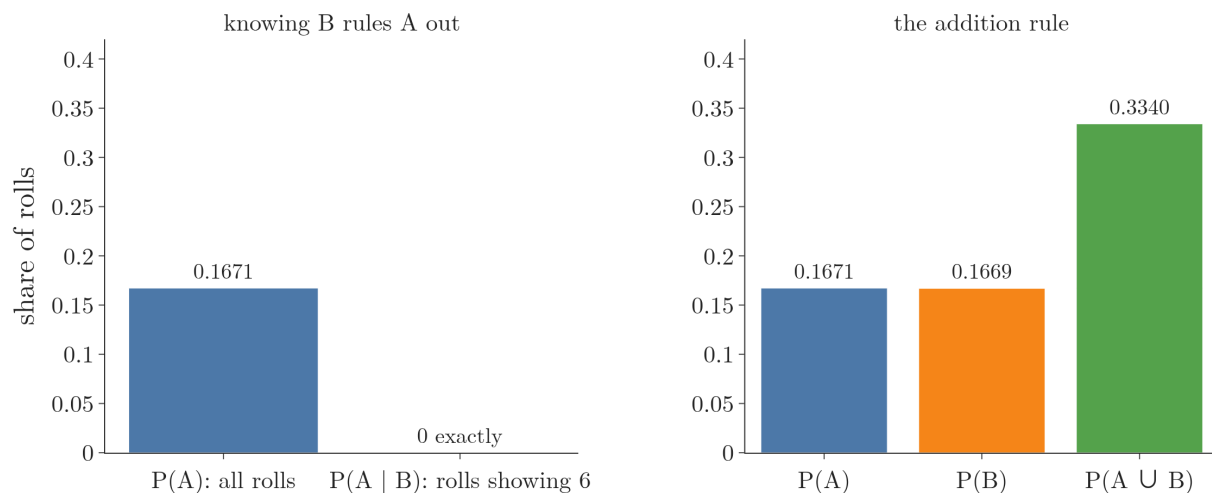


Figure 2: One million rolls of a fair die. Left: the share showing 3 is 0.1667 over all rolls, and exactly 0 among the rolls that showed 6. Right: the shares of 3, of 6, and of “3 or 6”, which is their sum, about $1/3$.

4. Mutually exclusive is not independent

Key point: Knowing B happened tells us A did not, which is a lot of information. So mutually exclusive events (each with a positive probability) are always dependent.

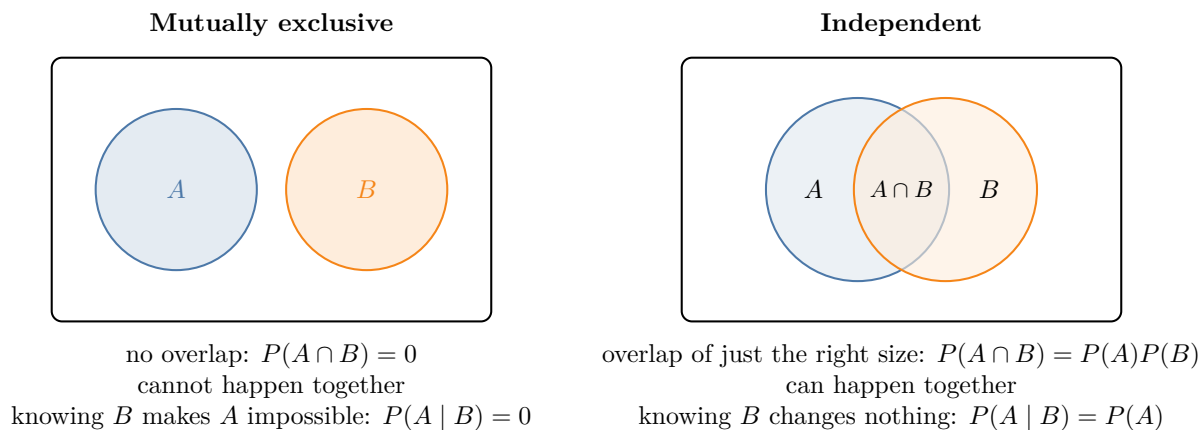


Figure 3: Mutually exclusive versus independent events

Figure 3 lays the two ideas side by side.

	Mutually exclusive	Independent
Can happen together?	no	yes
$P(A \cap B)$	0	$P(A) \times P(B)$
$P(A B)$	0	$P(A)$
Does B tell us about A?	yes: A did not happen	no

For “die 1 shows 3” (A) and “die 1 shows 6” (B), each has probability $1/6$:

- $P(A \cap B) = 0$, so they are mutually exclusive.
- $P(A) \times P(B) = 1/36 \neq 0$, so they are **not** independent.

In fact, if both events have a probability above 0, being mutually exclusive always rules out being independent. Learning that one happened tells us for certain that the other did not.

An everyday picture: a light switch is either on or off. The two states are mutually exclusive, and for that very reason, hearing “the switch is on” tells us everything about “off”.

5. The addition rule: where mutual exclusivity pays off

Key point: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. For mutually exclusive events the last term is 0, so “or” becomes plain addition.

The probability that A **or** B happens (one, the other, or both) is the probability of their **union** (G-2045), written $P(A \cup B)$. Counting it needs care when the events share outcomes.

A bag holds 29 objects: 5 yellow cubes, 7 yellow spheres, 8 green cubes and 9 green spheres. One object is drawn at random. What is the probability that it is yellow or a cube?

1. **Count yellow:** $5 + 7 = 12$ objects, so $P(\text{yellow}) = 12/29$.
2. **Count cubes:** $5 + 8 = 13$ objects, so $P(\text{cube}) = 13/29$.
3. **Adding them counts some objects twice:** $12 + 13 = 25$, but the 5 yellow cubes are in both counts.
4. **Subtract the overlap once:** $12 + 13 - 5 = 20$, so $P(\text{yellow or cube}) = 20/29$.

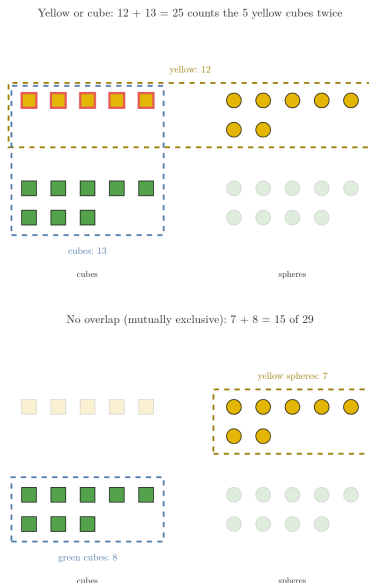


Figure 4: The bag of 29 objects. Yellow (12) and cube (13) share the 5 yellow cubes, so their counts add up to 25 with 5 counted twice; subtracting the 5 gives 20 of 29. Yellow spheres (7) and green cubes (8) share nothing, so their counts simply add: 15 of 29.

In Figure 4, watch the 5 yellow cubes (ringed in red): they lie inside both dashed boxes, so a plain sum counts them twice. Dividing every count in step 4 by 29 gives the general **addition rule**:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Now take two events with no overlap: “yellow sphere” (7 objects) and “green cube” (8 objects). No object is both, so the events are mutually exclusive and $P(A \cap B) = 0$. Nothing is counted twice (Figure 4, last scene), and the rule becomes a simple sum, the **addition rule for mutually exclusive events** (G-173):

$$P(A \cup B) = P(A) + P(B) = \frac{7}{29} + \frac{8}{29} = \frac{15}{29}$$

The die of Section 3 follows the same rule: $P(3 \text{ or } 6) = 1/6 + 1/6 = 1/3$ (Figure 2, right). The Bayes’ theorem Notes use this sum to split a probability into separate cases.

6. Summary

- Mutually exclusive: $P(A \cap B) = 0$; the events cannot happen together, and $P(A | B) = 0$.
- Independent: $P(A \cap B) = P(A)P(B)$; they can happen together, and $P(A | B) = P(A)$.
- Mutually exclusive events with positive probabilities are never independent.
- Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; for mutually exclusive events, $P(A \cup B) = P(A) + P(B)$.

7. Sources

Built from

- CampusX, “Naive Bayes Classifier | Part 3 | Mutually Exclusive Events”, YouTube, <https://www.youtube.com/watch?v=nneTjTYikBE>
- Khan Academy, “Addition rule for probability | Probability and Statistics”, YouTube, <https://www.youtube.com/watch?v=QE2uR6Z-NcU> (the bag of 29 objects and the addition rule; Figure 4 recreates the count with our own drawing)

8. Key terms

Term	Meaning
Mutually exclusive events	Events that cannot happen at the same time; their intersection has probability 0
Union ($A \cup B$)	The event that A or B (or both) happens
Addition rule (G-173)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$; for mutually exclusive events, $P(A \cup B) = P(A) + P(B)$